ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B. Tech. (ME)

Semester: VI

SUBJECT CODE: BOE602

ADDITIVE MANUFACTURING TECHNOLOGY

Time: 03:00 Hrs. **Maximum Marks: 70**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following questions. (14 Marks)

Q. 1. Define the following terms: (1×7 = 7 Marks)

(i) Additive Manufacturing (AM)
(ii) LOM
(iii) SLM
(iv) Rapid Prototyping
(v) Toolpath Planning
(vi) Staircase Effect
(vii) Electroplating

Q. 2. Answer the following questions: (1×7 = 7 Marks)

(i) Give any two examples of subtractive manufacturing.
(ii) List any two CAD software that can be used in AM.
(iii) Write the full form of FDM.
(iv) Write any two area filling methods.
(v) STL stands for Standard Technical Language. (True/False)
(vi) Reverse engineering is used for existing products. (True/False)
(vii) SLA is the oldest AM technique. (True/False)

SECTION – B

NOTE:Attempt all questions. (7×2×2 = 28 Marks)

Q.3.(i)(a) Briefly explain the role of Additive Manufacturing technology in medical applications.

OR

(b) Classify Additive Manufacturing processes on the basis of material used.

(ii)(a) What are the various advantages and disadvantages of Additive Manufacturing over conventional approaches?

OR

(b) Briefly discuss the economic analysis of Additive Manufacturing process.

Q.4.(i)(a) Briefly explain various post processing operations of Additive Manufacturing.

OR

(b) How is the manipulation of STL files carried out?

(ii)(a) What are the possible errors in an additively manufactures part?

OR

(b) Differentiate between forward and reverse engineering.

SECTION – C

NOTE: Attempt all questions. (14×2 = 28 Marks)

Q.5.(a) Draw the set-up of SLA and discuss its working in detail.

OR

(b) Discuss the different types of slicing methods with suitable diagrams.

Q.6.(a) What are the different steps involved in Additive Manufacturing process? Explain each of them.

OR

(b) Explain the construction and working of SLS process of Additive Manufacturing.